Crebos Documentation

Daily Delivery Health Check System – Full Technical Documentation

AI & Workflow Automation Pilot – Crebos Online Solutions https://crebos.online/

watch demo: https://www.youtube.com/watch?v=c8ETK5kOh08

1. Purpose of the System

The Daily Delivery Health Check workflow automates monitoring of ongoing projects inside ClickUp and provides:

- **Objective scoring** (RED, YELLOW, GREEN)
- **AI-generated key issues and recommendations**
- **ClickUp task updates**
- **Escalation routing** for risky or unclear projects
- **Consolidated alerts** (so the ops team doesn't get spammed)
- **Fail-safe manual review loop** for malformed AI or ambiguous statuses

This system is built to run **daily**, fully hands-off.

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2. Architecture Overview

The workflow is built on **three synchronized phases**, each with clear responsibility separation:

Phase I: Data Preparation & Task Update (Core Processing)

Tasks are fetched, normalized, scored, enriched by AI, validated, and updated before any alerts are sent.

Key Operations in this Phase:

1. **ClickUp: Get Tasks**
   - Retrieves all project tasks from the target list.
   - Includes custom fields, blocker descriptions, progress, deadlines.
2. **Code: Normalize Project Data**
   - Converts all inputs into a clean, predictable schema.
   - Converts percentages to integers.
   - Ensures strings like `"null"` become actual null.
   - Computes **deadline pressure** (days left).
   - Eliminates undefined values that break AI.
3. **Code: Scoring Engine**
   - Implements deterministic cascaded logic:
     - If any Red rule fires → Status = Red.
     - Else if any Yellow rule fires → Status = Yellow.
     - Else → Green.
   - Ensures consistency regardless of inconsistent ClickUp fields.
4. **Gemini AI Enrichment**
   - Produces:
     - `aiKeyIssues`
     - `aiRecommendation`
     - `aiScoreValidation` (LLM cross-check)
   - Auto-retries 3 times.
   - Strict JSON output enforcement.
5. **Code: Validate AI Output**
   - If any of the required fields are missing, empty, malformed, or contain hallucination artifacts → route to **Manual Review**.

- Ensures workflow doesn’t break when AI misbehaves.

6. **Loop + ClickUp Update (Batch)**
   - Each task gets updated with:
     - Score
     - Status Color
     - AI Valid (boolean)
     - Recommendation
     - Key Issues
     - Last Checked timestamp
7. **Wait Node (3 seconds)**
   - Allows ClickUp backend to commit all updates before next phase.
   - Fixes inconsistencies where updates appear in API but not UI instantly.

Phase II: Verification & Routing Logic

This phase classifies tasks into two types:

1. Status Alerts

Tasks that:

- have valid AI,
- and status is **Red** or **Yellow**.

2. Manual Reviews

Tasks that:

- have invalid AI output,
- or missing core fields,
- or suspiciously inconsistent data.

Nodes in This Phase:

1. **ClickUp Get Tasks (Fetch Updated State)**
   - Pulls fresh state post-update.
   - Confirms fields are committed.
2. **Code: Manual Review Detection**

   For each task, generates:

   ```
   {
     "taskId": "...",
     "taskName": "...",
     "score": 25,
     "status": "Red",
     "blockers": "...",
     "aiValid": true,
     "notificationType": "STATUS_ALERT" | "MANUAL_REVIEW"
   }
   ```

3. **Code: Aggregate Alerts (Anti-Spam Fix)**
   - This is the **core architectural fix**.
   - Instead of n8n emitting 30 items (one per task):
     - It emits **one single item** containing two arrays:
       - `statusAlerts: []`
       - `manualReviews: []`
     - Also includes:
       - `statusAlertCount`
       - `manualReviewCount`

This change eliminated the 36+ spam emails problem.

Phase III: Notification, Manual Task Creation & Completion

This is the output/communication layer.

Nodes:

1. **IF: Status Alerts Exist?**
   - Checks `statusAlertCount > 0`
   - If true → Generate email.
2. **IF: Manual Reviews Exist?**
   - Checks `manualReviewCount > 0`
   - If true:
     - Creates a new **Manual Review ClickUp Task**
     - Sends manual review email summary
3. **Gmail: Status Alert Email**

   Format includes:
   - Project name
   - Score
   - Blockers
   - Key Issues (AI)
   - AI Recommendations (AI)
   - Link to task
4. **Gmail: Manual Review Email**

   Summary of all projects requiring human evaluation.
5. **Merge Node (Wait for All Inputs)**
   - Ensures completion only logs after all alert tasks are sent.
6. **Final Comment / Final Log (Optional)**
   - Marks the run complete.

3. Key Engineering Decisions & Trade-Offs

Decision	Why it matters	Trade-off
Aggregation before routing	Prevented 36 duplicate emails. Clean architecture.	Required rethinking of pipeline.
Formally validated AI output	AI never breaks downstream nodes.	Additional code node.
Two-pass ClickUp fetch	Guarantees accurate verification after updates.	Slightly longer run time.
Split into Alert vs Manual Review	Clearer oversight & escalation.	More branches & nodes.
Wait node before re-fetching	Ensures ClickUp commits custom field updates.	Adds ~3 seconds per run.
Loop-based update	Guarantees updates even on variable data sizes.	More nodes in update pipeline.
Strict fallback to manual review	Zero silent failures.	Requires human intervention for malformed AI.

4. Full Error Handling Strategy

The system gracefully handles:

- AI malformed JSON
- Missing fields
- Null blockers
- Bad percentages
- Undefined ClickUp fields
- Network delays
- ClickUp caching delays
- Duplicate alerts
- Duplicate comments

- Partial task updates

Recovery Methods:

- AI retry mechanism
- Normalizer resets invalid values
- Manual review fallback
- Aggregated routing
- Type-checking in all Code nodes
- Timestamped audit log
- Fail-soft instead of fail-hard design

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5. Anti-Duplicate & Anti-Spam Controls

1. Email Duplication Fix

Now sends exactly:

- 1 status alert email
- 1 manual review email

  (only if relevant)

2. Duplicate comment prevention

A comment signature prevents repeated daily comments per task.

3. Prevent reprocessing

Tasks processed today are tagged with a fresh timestamp.

4. Consolidation logic

The `Aggregate Alerts` node collapses 30 items → 1.

This is what makes the system scale to 100+ tasks.

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6. Output Examples (Clean & Structured)

Status Alert Email Output

Contains:

- Task Name
- Status Color
- Score
- Blockers
- AI Key Issues
- AI Recommendations
- Link to task

Manual Review Email Output

Contains:

- A table-style summary of all tasks requiring review
- Reason for manual review (missing fields / invalid AI)
- Confirmation that a ClickUp “Manual Review Task” was created

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7. Node Inventory (Simplified)

Total nodes: 25+

Core Node Types Used:

- Trigger
- ClickUp (Get / Update / Create)
- Code (Normalize, Score, Validate, Aggregate)
- Gemini (AI)
- IF (Routing)

- Merge (Synchronize paths)
- Wait (Stabilization)
- Gmail (Notifications)
- Loop (Batch updates)

This is a *serious* pipeline, not a toy workflow.

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8. Final Summary of system:

- Fault tolerance
- AI augmentation
- Data validation
- Anti-spam routing
- Real-time task updates
- Clear escalation logic
- Full reporting
- End-to-end stability

AI Prompt used HTTP body using Google Gemini 2.5 flash via API key

You are a project health analyst. Analyze this project and provide actionable insights in JSON format ONLY.

Project: {{projectName}}
Completion: {{completion}}%
Overdue Tasks: {{overdue}}
Blockers: {{blockers}}
My Assessment: Status is {{systemStatus}} with score {{score}}/100

RESPOND WITH ONLY THIS JSON (no markdown, no extra text):
{
"aiStatus": "RED" | "YELLOW" | "GREEN",

"isValid": true | false,
"keyIssues": "2-3 key problems (2 sentences max)",
"risks": "What happens if not fixed (2 sentences max)",
"recommendedActions": "Specific next steps (2-3 actions)"
}

Rules:

- aiStatus MUST match the system-calculated status ({{systemStatus}})
- isValid = true if you agree with the system, false if your assessment differs
- Be concise and actionable
- Return ONLY valid JSON, nothing else.